DIGITAL TWIN CHALLENGE 2026 · HAPPIEST HEALTH

CardioTwin

A hypertension digital twin that simulates interventions, not just risk

315M
INDIAN ADULTS WITH HYPERTENSION
ICMR-INDIAB, LANCET D&E 2023

0.960
AUROC DETECTING THE NON-DIPPER PHENOTYPE WITH NO BP CUFF

5
INTERVENTIONS SIMULATED AND VALIDATED BY PAIRED RE-SIMULATION

Every patient in this work is simulated. No real patient data is used. Research prototype, not a medical device.

THE PROBLEM

315 million Indians have hypertension. Almost none get the test that matters.

Blood pressure follows a circadian rhythm. It should fall 10–20% overnight. People whose pressure does not fall are **non-dippers**, and non-dipping predicts cardiovascular events independently of the daytime reading. Reverse dippers, whose pressure rises at night, are at higher risk still.

Confirming the phenotype requires 24-hour ambulatory blood-pressure monitoring. ABPM is scarce, costly and largely urban-private in India, so the overwhelming majority of the 315 million are managed on clinic readings that cannot see the thing that matters most.

<10%

NOCTURNAL FALL DEFINES
A NON-DIPPER

101M

INDIAN ADULTS WITH DIABETES,
A MAJOR DRIVER OF NON-DIPPING

~0

ROUTINE ABPM IN MOST
INDIAN PRIMARY CARE

Meanwhile the wrist is already instrumented. Consumer wearables measure heart rate, heart-rate variability, sleep architecture and activity continuously. They cannot measure blood pressure. The question this project asks is whether what they *can* measure is enough to find the people who need the cuff.

WHAT MAKES IT A TWIN

A predictor says what will happen. A twin says what happens if you change something.

We hold both to that standard. CardioTwin is a reduced mechanistic model of the patient's circadian pressure, personalised from one calibration week:

$$SBP(t) = S \cdot circ(t;\ night_f) + surge(t;\ A) + behaviour(t) - med(t) + \varepsilon$$

Setpoint **S**, nocturnal factor **$night_f$** and surge amplitude **A** are fitted per patient from their own trace. Behaviour coefficients are estimated once across the cohort by within-patient regression on observed data. The twin never reads the simulator's latent traits.

Because the structure is mechanistic rather than a black box, you can intervene on it: move a dose to bedtime, extend sleep, cut salt, raise activity, fix adherence, and re-integrate.

And because we own the ground-truth simulator, the what-if engine can be checked. For every intervention we re-run the same virtual patient with the same random seed and the same override, and compare. Paired, no cherry-picking.

DATA · TWO STREAMS, FULLY SYNTHETIC

A mechanistic cohort, not a sampled noise generator

Static: synthetic EHR

- Demographics, urban/rural, South-Asian adiposity
- ICD-10 problem list: I10, E11.9, N18.3, G47.33, I21.9
- Labs: creatinine, eGFR, HbA1c, LDL, Na, K, UACR
- Polygenic risk proxy, regimen with dose timing

Dynamic: wearable at 5-minute resolution

- Heart rate, RMSSD, sleep stage, steps, cuff BP
- 300 patients × 21 days = 1.81M samples
- Circadian rhythm with dipping phenotype
- Morning surge, autonomic coupling, drug PK

Four pieces of established physiology are encoded: the dipping rhythm, the morning surge, the nocturnal vagal rise that is blunted in non-dippers, and steady-state pharmacodynamics so a missed or badly timed dose is visible in the trace.

Face validity check. The simulator was never told the clinical cut-off, yet the phenotypes it produces land exactly where the definitions sit: dippers fall **15.6%** overnight, non-dippers **6.0%**, reverse dippers **rise 2.6%**.

RESULT 1 · THE NEGATIVE ONE

Cuff-free wearables do not improve next-day event forecasting

Task: predict a sustained Stage-2 excursion (SBP ≥ 160 for ≥ 30 min) in the next 24 hours. Patient-grouped split, 90 held-out patients, 1260 patient-days, prevalence 30%. Bootstrap CIs resampled by patient.

MODEL	AUROC	95% CI	AUPRC	BRIER
EHR only	0.637	[0.528–0.731]	0.473	0.198
Cuff-free wearable only	0.517	[0.433–0.599]	0.315	0.211
Fused, cuff-free	**0.642**	[0.550–0.725]	0.462	0.197
Reference: with BP cuff	0.882	[0.836–0.918]	0.835	0.107
Baseline: yesterday's mean SBP	0.884	[0.839–0.920]	—	—

We report this as a failure of the hypothesis. Fusing cuff-free wearable data with the record (0.642) does not beat the record alone (0.637); the intervals overlap almost entirely. If a cuff is available, use it (0.882) — and a one-line rule on yesterday's reading is already 0.884. Forecasting tomorrow's pressure from the wrist is not the product.

RESULT 2 · THE ONE THAT MATTERS

The wrist can find the non-dipper, and that is the clinically scarce test

Same cuff-free inputs. Different question: not *when will pressure spike* but *does this patient's pressure fail to fall at night*.

0.960

AUROC, PATIENT-LEVEL, CUFF-FREE
95% CI 0.911–0.998 · N = 90 HELD-OUT PATIENTS

The signal is autonomic. Non-dippers show a blunted overnight rise in RMSSD, so the night/day HRV ratio carries the phenotype even with no pressure measurement anywhere in the input.

Why this is the useful result

- Non-dipping independently predicts cardiovascular events
- Confirming it needs ABPM, which India largely does not have
- A wearable screen turns a scarce test into a triage decision: who actually needs the cuff
- It runs on hardware tens of millions of Indians already wear

The honest caveat. This result follows from an autonomic-dipping coupling we encoded in the simulator. The coupling is attested in the physiology literature; its effect size is our choice. This number is evidence that the pipeline works, not evidence about real patients. Real ABPM plus wearable data is the next step.

RESULT 3 · THE TWIN, MEASURED

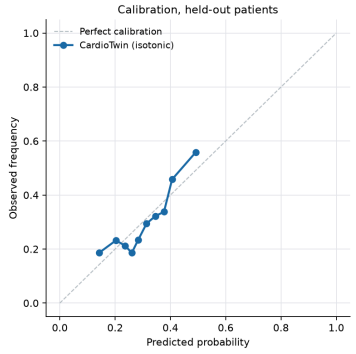
Counterfactual accuracy against paired re-simulation

For each intervention the twin predicts a change; the ground-truth simulator is re-run for the same virtual patient under the same seed. 120 patients, all five interventions, no selection.

INTERVENTION	TRUE Δ24H	TWIN Δ24H	MAE	TRUE ΔNIGHT	TWIN ΔNIGHT	SIGN
Increase activity by 30%	-0.64	-0.47	0.17	-0.96	-0.47	100%
Move one agent to 21:00 dosing	+0.10	+0.01	0.09	-1.75	-2.58	99%
Perfect medication adherence	-2.16	-2.31	1.53	-1.73	-1.90	90%
Reduce salt by 3 g/day	-2.85	-2.85	0.00	-2.85	-2.85	100%
Extend sleep by 1 hour	-0.50	-0.78	0.65	+0.03	-0.78	74%

Chronotherapy is the showcase. Moving one agent to 21:00 changes the 24-hour mean by essentially nothing (+0.10 mmHg) while lowering night-time pressure by 1.75 mmHg. A scalar risk score cannot express that. A twin can, and that is precisely the decision a clinician faces with a non-dipper.

Two arms we do not claim. The salt arm is `circular` — the twin's coefficient and the simulator's are both drawn from the same literature value, so it tests nothing. The sleep arm is `weak` at 74% sign agreement; the twin over-states the benefit because it over-amplifies debt carry-over.



RESULT 4 · FALSIFIABILITY

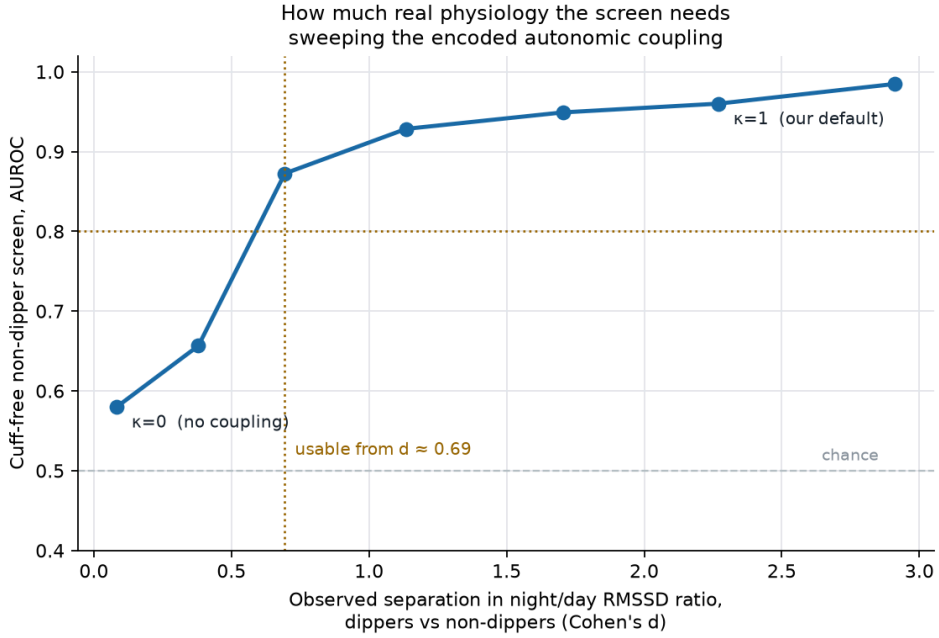
How much of this is just our own assumption?

The obvious attack on Result 2 is that we encoded the autonomic coupling and then detected it. So we measured exactly how much we assumed. We sweep the coupling strength, regenerate the whole cohort each time, retrain, and plot the screen against the **observed** separation between dippers and non-dippers, which is a quantity anyone can measure in real patients.

Negative control holds. At zero coupling the screen collapses to **0.580**, near chance. The model has no back-door route to the label. The small residual above 0.5 comes from sleep apnoea, which independently affects both dipping and HRV.

The threshold. The screen becomes usable (AUROC 0.80) once the observed separation reaches **d ≈ 0.694**. Our default assumes d = 2.27, which is almost certainly generous.

That turns an assumption into something falsifiable: **this approach works if, and only if, real non-dippers show a night/day RMSSD separation of at least d ≈ 0.694.** Anyone with paired ABPM and wearable data can check whether reality clears the bar. Result 6 is our attempt to start.



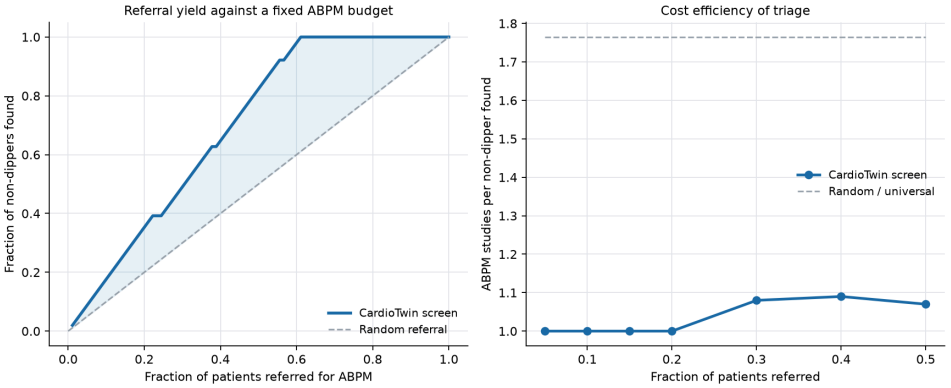
RESULT 5 · OPERATIONAL VALUE

An AUROC is not a decision. Who gets the ABPM?

BUDGET	STUDIES	FOUND	PRECISION	SENSITIVITY	ABPM/FIND
5%	4	4	100%	8%	1.00
10%	9	9	100%	18%	1.00
15%	14	14	100%	27%	1.00
20%	18	18	100%	35%	1.00
30%	27	25	93%	49%	1.08
40%	36	33	92%	65%	1.09
50%	45	42	93%	82%	1.07
universal	90	51	57%	100%	1.76

At a 20% referral budget every single referral is a true non-dipper (100% precision against 57% for random referral), catching 35% of all non-dippers for a third of the studies.

Honest ceiling. At 57% prevalence, triage can lift yield by at most 1.76×. The value here is in not wasting studies, not in finding something rare. A lower-prevalence screening population would gain more.



RESULT 6 · REAL PATIENTS

External validation on data we did not generate

SHAREE (Melillo et al., PLoS ONE 2015): 139 real hypertensive patients, nominal 24-hour Holter at 128 Hz, 12-month follow-up, 17 vascular events (11 myocardial infarction, 3 stroke, 3 syncope). Open access, no credentialing.

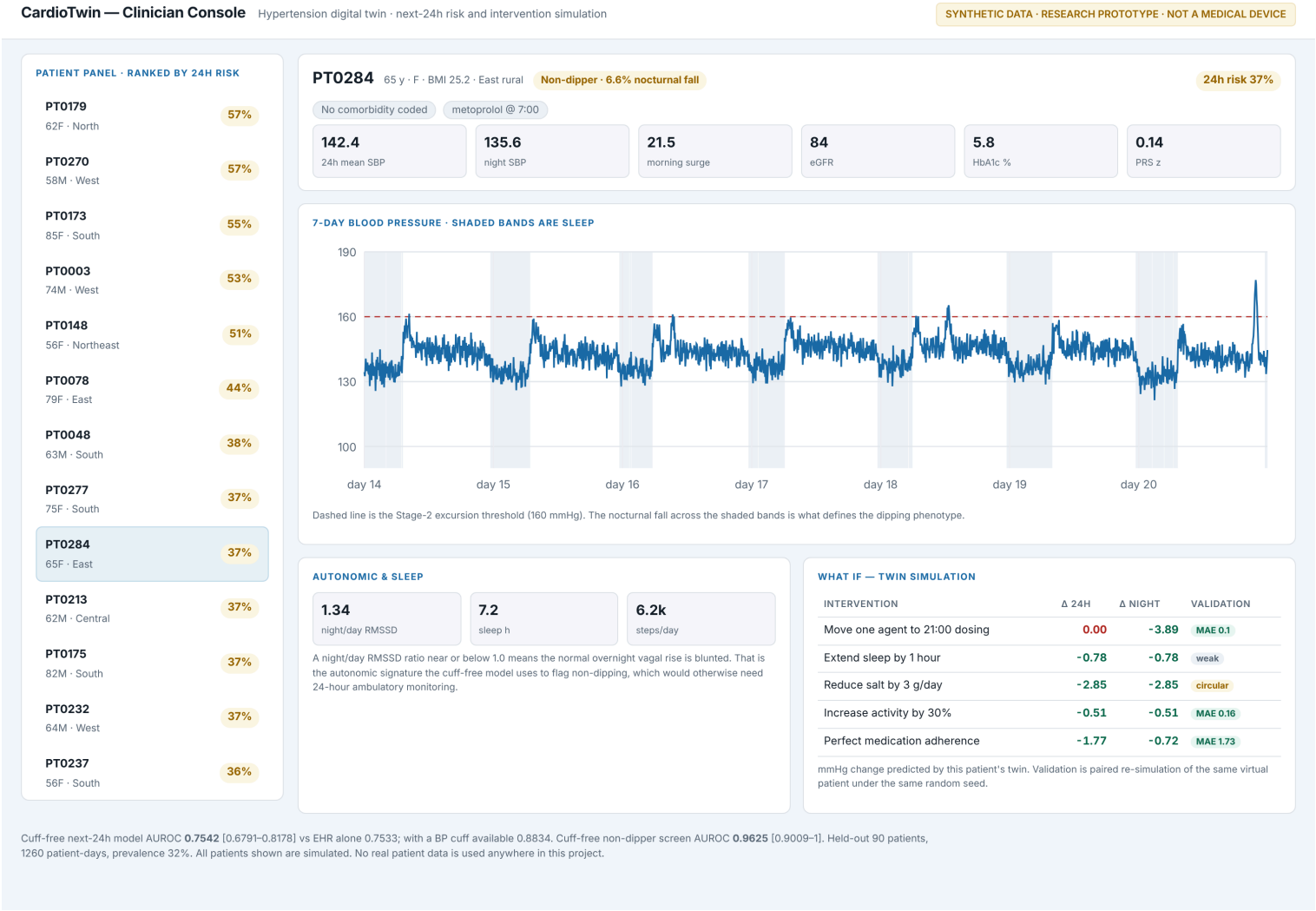
The pipeline is written and in the repository (`src/validate_sharee.py`). It extracts RR intervals from the shipped QRS annotations, computes night and day HRV, and tests whether the night/day RMSSD ratio predicts a 12-month vascular event over and above age, sex and clinic blood pressure.

Status: analysis pending at the time of writing. We are reporting that plainly rather than leaving the slide out. Whatever it returns, including a null, goes into the repository.

What it will and will not settle. It tests the prognostic value of the autonomic feature in real hypertensive patients. It cannot validate non-dipper detection itself, because SHAREE carries no ambulatory blood pressure, so the dipping phenotype is unobservable in it.

THE CONSOLE

What the doctor actually sees



LIMITATIONS

What this is not

Everything is in silico

- No real patient data anywhere in the project
- Performance reflects our simulator's assumptions, so it is an upper bound on what real data would give
- The non-dipper result depends on a coupling we encoded

Method limits

- Cuff-free forecasting adds nothing over the record (Result 1)
- Salt arm is circular; sleep arm over-predicts
- Twin is a reduced model; no baroreflex, renal or RAAS dynamics
- Cohort is enriched for non-dipping and is not population-representative

Clinical status. This is a research prototype. It is not a medical device, it has not been validated on humans, and nothing in it should be used to make a treatment decision. The dose-timing simulation in particular illustrates a mechanism; chronotherapy in real patients remains an area of active and unsettled trial evidence.

What would change our minds: paired ABPM and wearable recordings from an Indian hypertensive cohort. If the night/day RMSSD ratio does not separate dippers from non-dippers there, the screen does not work and we would say so.

WHERE IT GOES

From prototype to a triage layer that fits Indian primary care

Next

- Validate on open paired ABPM + wearable data
- Replace the reduced twin with a baroreflex model
- Calibrate on a real Indian cohort

Then

- Prospective pilot in an urban clinic
- ABPM referral driven by the screen
- Measure cuffs saved per non-dipper found

Deployment shape

- One calibration week with a cuff
- Continuous cuff-free monitoring after
- Escalate only when the twin flags

Find the patients whose pressure never sleeps, using the device they already wear.

Reproduce every number here with `./run_all.sh`. MIT licensed. All simulated data synthetic.